Coin Race pricer: vol diagnosis and fix

Bottom line

- The vol model works. Rebuilt end to end on Coinbase order-book data, it is calibrated on two held-out days. Its fair values match or beat the Kalshi market's log loss against settlement in every bucket from 1 to 15 minutes before close.
- The ETH under-pricing comes from the inputs the live pricer feeds the model. HYPE's sd is about 5× too large, ETH's about 2×, and BTC's about 1.5×.
- Rebuilt with correct vol, your logged tick (KXCRYPTOLEAD15M-26OCT080015 with 275 s left) prices ETH at 0.849, against a market mid of 0.842. ETH won.
- The HYPE series feeding your features and correlations is noisy or broken. A clean Coinbase HYPE series correlates 0.6–0.7 with BTC on 30–60 s returns, while yours shows 0.07–0.25.

Evidence

1. Reference pipeline vs Kalshi (Sep 1 and Oct 1 2026, held out of the HAR fit)

Data:

- Coinbase `book_snapshot_25` from Tardis, sampled every second.
- Kalshi 1-minute candles (bid/ask) and settled results.
- 151 events, 1,611 event-minute snapshots where all 5 markets had quotes.

The settlement check uses a Coinbase top-of-book 60 s TWAP. It picks the Kalshi winner 94.3% of the time on Oct 1 and 97.9% on Sep 1. Every miss was decided by less than about 1 bp.

Log loss against settlement (lower is better):

Time to close	n	Market mid	Reference (tv=5)	Ref + your 0.88/0.03 shrink	Your vol inflation	Inflation + your corr	No corr
≤1 min	60	0.383	**0.341**	0.432	0.549	0.581	0.449
1–3 min	114	0.660	**0.629**	0.674	0.932	0.955	0.834
3–7 min	436	1.074	**1.036**	1.053	1.213	1.251	1.195
7–15 min	1001	1.367	1.366	1.365	1.442	1.485	1.444
All	1611	1.201	**1.186**	1.197	1.311	1.350	1.296

- "Your vol inflation" multiplies the sds by the ratios measured at your tick: BTC 1.5, ETH 2.1, SOL 1.2, XRP 1.0, HYPE 5.3.

- The market's loss minus the reference's is 0.015 nats, with a bootstrap 90% CI of [−0.002, 0.034] by event.
- On each day separately: Sep 1 market 1.170 vs reference 1.149, and Oct 1 market 1.234 vs reference 1.226.

Calibration of the reference (mean predicted vs realized frequency):

Bin	0–5%	5–15%	15–30%	30–50%	50–70%	70–85%	85–100%
Reference	1.9/2.1	9.8/10.3	21.7/21.8	38.3/40.3	58.9/54.7	76.4/66.5	93.8/90.9
Market	3.0/2.1	9.4/9.3	21.3/22.3	38.4/39.6	58.8/56.6	76.6/70.8	90.6/96.7

HAR calibration on the held-out days, as mean(r^2)/mean(f) at h = 300 s with robust features:

Day	BTC	ETH	SOL	XRP	HYPE
Oct 1	0.83	0.86	0.89	0.87	0.98
Sep 1	0.88	0.85	0.87	0.89	0.91

Your live values were BTC 0.34, ETH 0.29, SOL 0.66, XRP 0.35, HYPE 0.06.

2. Your logged tick, rebuilt

Setup: 04:10:25 UTC on Oct 8, x taken from your log, HAR features approximated from Coinbase 1-minute candles, correlation from 4 hours of 1-minute returns.

	BTC	ETH	SOL	XRP	HYPE
Your sd (%)	0.147	0.236	0.141	0.155	0.639
sd implied by the trailing 1 h realized (%)	0.096	0.110	0.120	0.155	0.121
Reference HAR sd (%)	0.115	0.136	0.144	0.170	0.137
Fair, reference vol	0.048	**0.849**	0.000	0.009	0.094
Fair, your sd	0.134	0.479	0.003	0.014	0.371
Fair, your sd with HYPE fixed	0.124	0.709	0.002	0.011	0.154

The market mid was 0.842, and ETH settled yes.

3. Correlation is a symptom, not a parameter to tune

HYPE vs BTC correlation of Coinbase mid returns:

Data	1 s	10 s	30 s	60 s
Oct 1 (all coins)	0.40	0.56–0.58	0.61–0.63	0.62–0.64
Live recording, Oct 8, about 25 minutes	0.33–0.45	0.72–0.77	0.70	0.71

Adding idiosyncratic noise to a coin's series inflates its variance and dilutes its correlations together. A 16× variance inflation cuts correlation by about 4× (0.6 → 0.15), which matches what

you measured.

4. Which defect reproduces your symptoms (fault injection on Oct 1 HYPE)

Injected defect	Forecast inflation (raw 1 s features)	Inflation (robust 10 s features)	Forward 300 s r^2 ratio	Corr vs BTC, 60 s
Clean	1.0	1.0	1.00	0.62
IID noise, 2 bp	6.4	1.6	1.02	0.59
IID noise, 4 bp	20.2	3.1	1.07	0.52
Reverting 30 bp 1 s glitches (0.5% of seconds)	5.7	1.5	1.02	0.59
Reverting 30 bp 5 s glitches	5.7	3.3	1.10	0.52
Stale 30 s freezes	1.0	1.0	1.00	0.60
60 s one-sided-book jumps of 50 bp	2.5	2.1	1.42	0.37

High-frequency noise of about 3–4 bp on your HYPE series reproduces the roughly 16× forecast inflation while your forward 300 s check still looks clean. To get correlations as low as 0.1–0.2 at 60 s, the series also needs noise that persists for tens of seconds. That fits a composite flipping between price levels, for example when the utilized depth or size cap toggles, or a mix of products or venues.

Fix

1. **Find out what's wrong with the HYPE (and ETH) input series.** Run `python diagnose_bars.py your_bars.parquet` on your persisted bars. It needs columns `ts, coin, price` plus `mid` = raw Coinbase top-of-book mid if you have it. On a clean feed you should see:
   - signature RV(1)/RV(60) of 0.9–1.5,
   - |price − mid| rms of 1.5 bp or less,
   - HYPE vs BTC correlation of 0.55–0.75 at 30–60 s,
   - mean(r^2)/mean(f) of 0.85–1.0.

   Likely causes, in order: the CF-replica weighting jumping on thin HYPE depth, bar-loop artifacts (stalls, values from different seconds, duplicate or mixed sources), and the wrong product or a stale book after resubscribing.
2. **Compute HAR features and correlations from the raw Coinbase top-of-book mid, not the CF replica.** The mid matches settlement winners as well as the replica (93–98%), and it has no extra noise. If you keep the replica, use it only for x and the TWAPs.
3. **Use the robust features in `rv2.rv_features(p, k=10, clip=6)` with the matching `har_coefs.json`.** Squared 10 s returns are averaged over every 1 s offset, and each return is clipped at 6× a lagged EWMA sd. This is free on clean data (+0.2–0.3 held-out NLL gain) and much safer on bad ticks: a 30 bp glitch inflates the forecast 1.5× instead of 5.7×. The

coefficients are refit for h = 30/60/120/300/600/900 s, so interpolate the rate in log h at h = time remaining (see `pricer_ref.HAR.rate`). This also handles the horizon-rate wrinkle you raised.

4. **Treat the online scale c_coin as a monitor, not a fix.** Clamp it to [0.6, 1.6]. If its raw value leaves [0.4, 2.5], fire an alarm or kill switch. A value of 0.06 means the inputs are broken, and letting the scale absorb it would also hide the broken correlations. Drop the HYPE 1.13 factor.
5. **Remove the shrink.** Keep the t(ν=5) mixture. It is slightly better than a normal distribution from 1 to 7 minutes out, and about equal elsewhere. The cap costs 0.09 nats in the last minute, and wiring up `BaseRateTracker` would not change that.
6. **Correlation:** use a causal EWMA of 60 s returns (sampled every 10 s, 4 h half-life) on the mid; see `corr_series`. Don't hard-code it. Setting the correlation to zero costs 0.11 nats.
7. **Fast path:** precompute W once a second from the current horizon's covariance, then scale by $\sqrt{\tau_{\text{eff}}}$ ratios within the second. Compute bump deltas on the same draws, and enforce the theo-age kill switch.
8. **Test against** `golden_cases.json`. It holds 4 ticks (840/275/120/30 s left) built from `golden_mids_20261001_1000-1200utc.csv`. Each has log features, rates, sd, R and fair values (200k t-mixture draws; tolerance ±0.005). Your features and rates should match to about 1e-6.

Files

- `rv2.py`: robust RV features.
- `pricer_ref.py`: reference HAR, τ_eff, correlation EWMA and CRN Monte Carlo pricer.
- `har_coefs.json`: production HAR fit on all 22 days, (k = 10 s, clip 6).
- `diagnose_bars.py`: feed health check for your persisted bars.
- `golden_cases.json`, `golden_mids_*.csv`: regression vectors.
- `comp.py`: mid, microprice and an approximate single-venue CF Spot Rate.
- Validation scripts: `replicate.py` (settlement match), `validate.py`, `analyze.py` and `pooled.py` (Kalshi comparison), `calib.py`, `inject.py`, `har_fit.py`. `har_fit.py` and `calib.py` import `rvlib` from the earlier `rv_model/` package.

Caveats

- Validation covers two days (151 events). The edge over the market isn't statistically significant; what the data supports is that the theo is calibrated and competitive.
- The market beats the theo in the last minute on Oct 1 (19 snapshots), but not on Sep 1.
- The tick rebuild approximates HAR features from 1-minute candles because the live recorder started after that tick.